

In this Retail Customer Retention Analytics Project, TESCO provides five datasets: Customer Demographics, Customer Transactions, Store Locations, Loyalty Program, and Churn Labelled Customers. I will analyse these datasets, create visuals, and derive insights for TESCO.

Explanation Video :

<https://drive.google.com/file/d/1krrqdZtEW2qPGIEjFuE0DdsibloFRkSK/view?usp=sharing>

Task 1 Data Modelling and Cleaning

First, I downloaded these datasets and loaded them into Power BI Desktop.

➤ Use Power Query Editor to load and transform the datasets.

○ Remove duplicate rows based on key columns (e.g., Customer_ID, Store_ID)

I have imported all the datasets into the Power Query Editor. For each query, I applied the Remove Duplicates function to the key columns as follows:

- Customer Demographics - Remove Duplicate for Customer_ID
- Customer Transactions - Remove Duplicate for Transaction_ID
- Store Locations - Remove Duplicate for Store_ID
- Loyalty Program - Remove Duplicate for Customer_ID
- Churn Labelled Customers - Remove Duplicates for Customer_ID

○ Format columns: Convert dates, ensure numeric types for Amount, Points, etc.

I formatted the columns using the Change Data Type function, specifying the following:

Age as a Whole Number,

Membership_Since as a Date,

Membership_Duration as a Whole Number,

Amount as a Decimal Number, and

Transaction_Date as a Date, among others.

○ Handle missing or null values appropriately (e.g., filter or replace)

I have filtered each dataset and each column using the Filter function.

➤ Create calculated columns:

○ Membership_Duration = Today - Membership_Since

I created a calculated column in the Customer Demographics table using the following DAX formula:

```
Membership_Duration = DATEDIFF(Customer_Demographics[Membership_Since], TODAY(), DAY)
```

In this formula, I used the DATEDIFF function to calculate the number of days between the membership start date and the current date(Today).

- **Add a Transaction_Year and Transaction_Month column from Transaction_Date**

I created a calculated column as Transaction_Month in the Customer Transaction table using the following DAX formula:

```
MONTH(Customer_Transactions[Transaction_Date])
```

In this formula, I used the MONTH function to extract months from the date column.

I created a calculated column, Transaction_Year, in the Customer Transaction table using the following DAX formula:

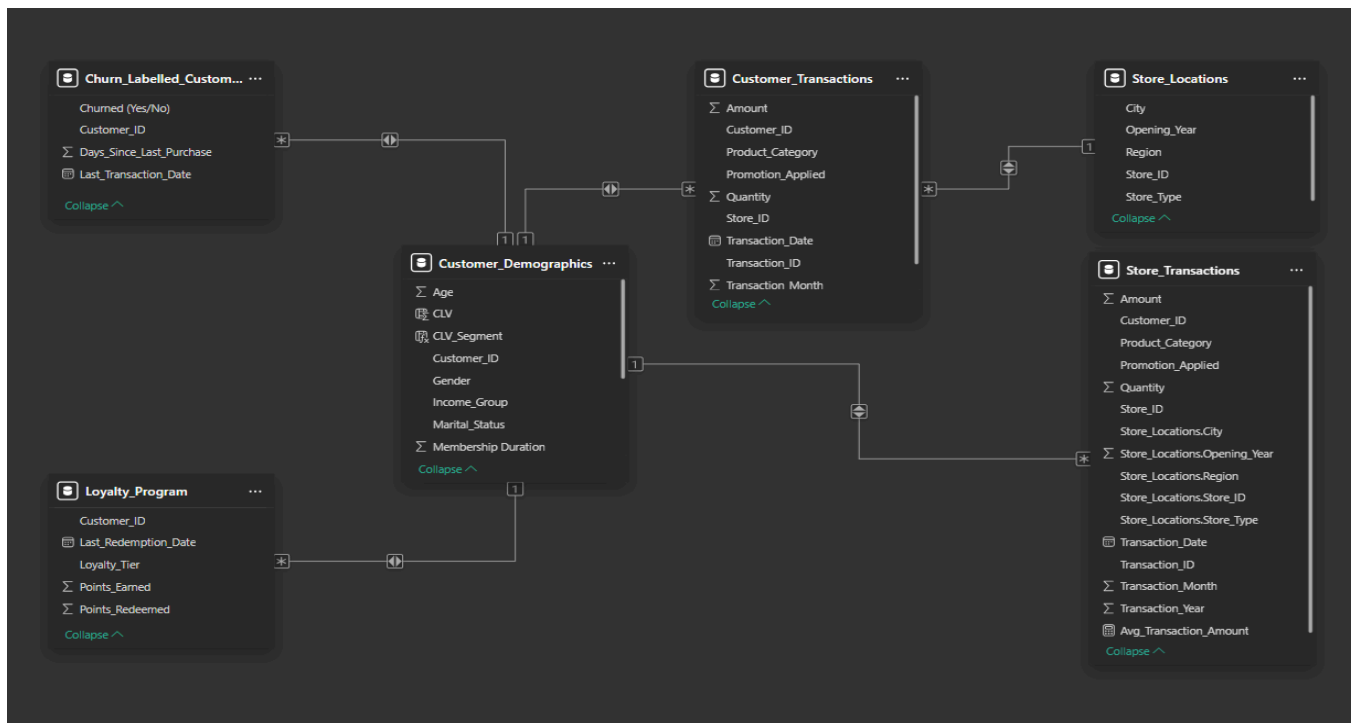
```
YEAR(Customer_Transactions[Transaction_Date])
```

In this formula, I used the YEAR function to extract years from the date column.

➤ **Create a basic Data Model view**

- **One-to-Many:** Customer_Demographics → Transactions, Loyalty_Program, Churn_Labelled_Customers
- **Many-to-One:** Transactions → Store_Locations

I have created relationships between tables, as we can see in the image:



Task 2 Churn and Retention Metrics

➤ Create a Churn Rate card: $(\text{Churned Customers} / \text{Total Customers}) * 100$

To calculate the churned customers, I created a measure using this DAX formula:

```
Churned Customers =  
CALCULATE(DISTINCTCOUNT(Customer_Demographics[Customer_ID]), (Churn_Labelle  
d_Customers[Churned (Yes/No)]="Yes"))
```

This measure counts how many unique customers have churned.

- `DISTINCTCOUNT(Customer_ID)` is used so each customer is only counted once.
- `CALCULATE` lets us apply a filter where `Churned = "Yes"`, so only churned customers are included.

Why: We use `CALCULATE` to change the context of the calculation and `DISTINCTCOUNT` to avoid double-counting customers.

Next, I need to calculate the total number of customers, so I created a measure using this DAX formula:

```
Total Customers = DISTINCTCOUNT(Customer_Demographics[Customer_ID])
```

This measure returns the total number of unique customers.

- `DISTINCTCOUNT(Customer_ID)` is used so each customer is counted only once, even if they appear multiple times in the data.

Why: We use `DISTINCTCOUNT` to avoid duplicate counts and get the true customer base size.

For Churn Rate, I have created a measure using this DAX formula:

```
Churn Rate % = ('Measuers Table'[Churned Customers]/'Measuers Table'[Total  
Customers]*100)
```

This measure calculates the percentage of customers who have churned.

- It divides Churned Customers by Total Customers, then multiplies by 100 to convert the result into a percentage.

Why: This gives a clear view of what portion of the total customer base has churned.

➤ **Visualise churn rate by:**

○ **Region**

To display and compare the Churn Rate by Region, I created a stacked column chart.

○ **Income Group**

To compare the Churn Rate across different Income Groups, I created a stacked bar chart.

○ **Store Type**

To compare the Churn Rate by Store Type, I created a stacked column chart.

○ **Identify the top 5 segments with the highest churn %**

I have created a text box that lists the top 5 segments with the highest churn rates.

➤ **Create funnel chart: Total Customers → Repeat → Churned**

To create a funnel chart, I have the Total Customers measure and the Churned Customers measure, so I need to calculate the Repeat Customers. So, I created a measure using the following DAX formula:

```
Repeat Customers =  
CALCULATE(DISTINCTCOUNT(Customer_Transactions[Customer_ID]),FILTER(VALUES(  
Customer_Transactions[Customer_ID]),CALCULATE(COUNTROWS(Customer_Transactions)) > 1))
```

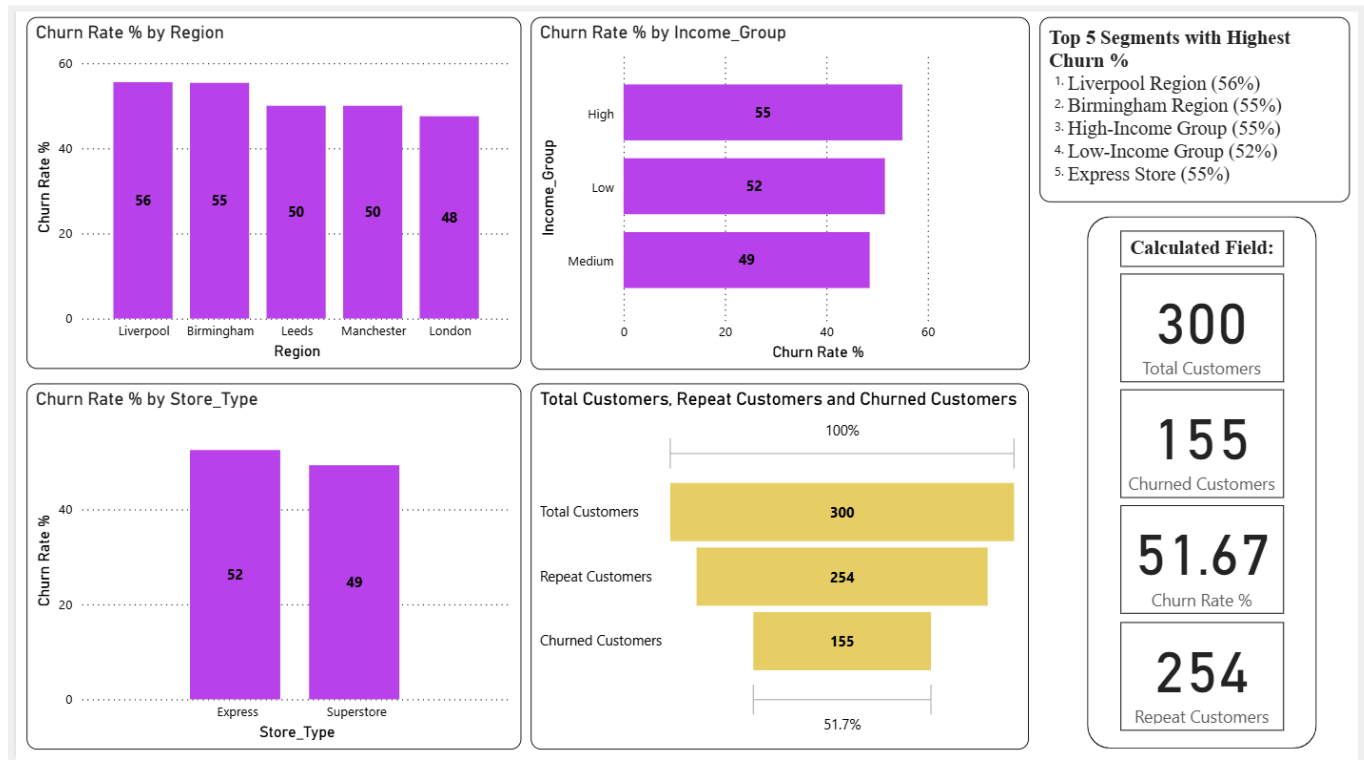
This measure counts how many unique customers have made more than one transaction.

- `COUNTROWS(Customer_Transactions) > 1` checks if a customer has more than one transaction.
- `FILTER + VALUES(Customer_ID)` applies this condition to each customer.
- `DISTINCTCOUNT(Customer_ID)` then counts only those customers who qualify.
- `CALCULATE` brings all these pieces together to apply the filter and return the result.

Why: We use this logic to identify and count customers who are repeat buyers instead of just one-time purchasers.

Afterwards, I created a funnel chart with Total Customers, Repeat customers, and Churned customers in that order.

This is task 2 Report Page view. As we can see in the image:



Interpretation of the page:

★ KPIs Cards:

- Total Customers: 300
- Repeat Customers: 254
- Churned Customers: 155
- Churn Rate: 51.7%

★ Churn by Region

- Highest churn: Liverpool (56%) and Birmingham (55%).
 - Lowest churn: London (48%).
 - Leeds and Manchester sit in the middle (50%).
- Customers in Liverpool and Birmingham are at significantly higher risk of leaving.

★ Churn by Income Group

- High-Income customers churn the most (55%), followed closely by Low-Income (52%).
 - Medium-Income customers churn less (49%).
- Surprisingly, even high-income customers are not retained well.

★ **Churn by Store Type**

- Express stores have higher churn (52%) than Superstores (49%).
Smaller, quicker-format stores are struggling more with retention.

★ **Top 5 High-Risk Segments**

1. Liverpool Region (56%)
2. Birmingham Region (55%)
3. High-Income Group (55%)
4. Low-Income Group (52%)
5. Express Stores (52%)

These represent the customer segments most likely to leave, and should be prioritised for retention strategies.

Recommendation for TESCO:

1. Target Liverpool & Birmingham with retention campaigns (loyalty programs, personalised offers).
2. Improve experience in Express stores – maybe quicker service or better product availability.
3. Focus on High-Income churn – provide premium perks or exclusive benefits to stop attrition.

Task 3 Repeat Purchase Analysis

➤ **Create a measure:**

Low-Tier Customers: (2-4 purchases)

Mid-Tier Customers: (5-10 purchases)

High-Tier Customers: (11+ purchases)

To implement these measures, I created two columns in the Customer Demographics table. The first column is called "Purchase Count," which indicates the number of times each customer has made a purchase. The second column is titled "Purchase Tier," which categorises customers into tiers based on the conditions mentioned above.

To calculate the Purchase Count for each customer, I created a new column using this DAX formula:

```
Purchase_Count =  
VAR Purchases =  
    CALCULATE(  
        COUNTROWS(Customer_Transactions),  
        FILTER(Customer_Transactions, Customer_Transactions[Customer_ID] =  
            Customer_Demographics[Customer_ID]))  
RETURN  
IF(ISBLANK(Purchases), 0, Purchases)
```

This calculated column shows how many transactions each customer has made.

- **COUNTROWS(Customer_Transactions)** counts the number of rows (transactions).
- **FILTER** ensures only transactions that match the current customer's **Customer_ID** are counted.
- **VAR Purchases** stores this count.
- **IF(ISBLANK(Purchases), 0, Purchases)** replaces blanks with 0 for customers who have no transactions.

Why: We use this to assign each customer their total purchase count, which helps categorise customers into tiers.

To categorise customers into tiers, I created a new column called Purchase Tier using a DAX formula.

```
Purchase_Tier =  
SWITCH(  
    TRUE(),  
    Customer_Demographics[Purchase_Count] = 0, "New",
```

```
Customer_Demographics[Purchase_Count] = 1, "One-time",
Customer_Demographics[Purchase_Count] >= 2 &&
Customer_Demographics[Purchase_Count] <= 4, "Low",
Customer_Demographics[Purchase_Count] >= 5 &&
Customer_Demographics[Purchase_Count] <= 10, "Mid",
Customer_Demographics[Purchase_Count] >= 11, "High"
)
```

This calculated column classifies customers into tiers based on how many purchases they've made (using Purchase_Count).

- 0 purchases → "New"
- 1 purchase → "One-time"
- 2–4 purchases → "Low"
- 5–10 purchases → "Mid"
- 11+ purchases → "High"
- The SWITCH(TRUE(), ...) structure makes it easy to evaluate multiple conditions in order.

Why: We categorise customers into tiers based on their purchasing frequency, which aids in retention and loyalty.

As we can see, both the calculated columns in the image below:

1 Purchase_Tier =										
Customer_ID	Gender	Age	Membership_Since	Marital_Status	Region	Income_Group	Membership_Duration	Purchase_Count	Purchase_Tier	
C1000	Male	50	01 November 2020	Single	London	High	1776	3	Low	
C1001	Female	18	05 July 2021	Divorced	London	Medium	1530	3	Low	
C1002	Male	36	18 August 2021	Single	Birmingham	Medium	1486	5	Mid	
C1003	Male	19	01 February 2024	Married	Leeds	Medium	589	3	Low	
C1004	Male	70	15 October 2020	Married	Leeds	Medium	1793	5	Mid	
C1005	Female	61	19 November 2020	Single	Liverpool	High	1758	0	New	
C1006	Male	43	13 July 2021	Divorced	London	High	1522	6	Mid	
C1007	Male	49	16 March 2023	Married	London	Medium	911	5	Mid	
C1008	Male	23	14 June 2022	Married	Manchester	Medium	1186	5	Mid	
C1009	Female	49	06 June 2024	Married	Liverpool	High	463	4	Low	
C1010	Male	21	03 May 2023	Single	Manchester	Medium	863	4	Low	
C1011	Male	28	07 July 2021	Married	Manchester	Low	1528	11	High	
C1012	Male	34	27 February 2023	Divorced	Manchester	Medium	928	2	Low	
C1013	Male	55	27 April 2023	Married	Birmingham	Medium	869	6	Mid	
C1014	Female	41	20 January 2024	Married	Birmingham	Low	601	3	Low	
C1015	Male	22	15 November 2023	Divorced	Manchester	Low	667	2	Low	

Afterwards, I created the measures as follows:

1. For Low Tier customers, I used this DAX formula:

```
Low Tier Customers =  
CALCULATE(COUNT(Customer_Demographics[Purchase_Tier]),Customer_Demographic  
s[Purchase_Tier] = "Low")
```

This measure counts the number of customers who fall into the “Low” purchase tier.

- COUNT(Purchase_Tier) counts the rows where a purchase tier is assigned.
- CALCULATE(..., Purchase_Tier = "Low") applies a filter so only customers with the “Low” tier are included.

Why: We use this to quickly see how many customers belong to the low purchase activity segment.

2. For Mid Tier customers, I used this DAX formula:

```
Mid Tier Customers =  
CALCULATE(COUNT(Customer_Demographics[Purchase_Tier]),Customer_Demographic  
s[Purchase_Tier] = "Mid")
```

Why: We use this to quickly see how many customers belong to the mid purchase activity segment.

3. For High Tier customers, I used this DAX formula:

```
High Tier Customers =  
CALCULATE(COUNT(Customer_Demographics[Purchase_Tier]),Customer_Demographic  
s[Purchase_Tier] = "High")
```

Why: We use this to quickly see how many customers belong to the high purchase activity segment.

➤ **Compare avg. purchase frequency by:**

Firstly, I need to create a measure called Average Purchase Frequency for comparison. I will use this DAX formula:

```
Avg Purchase Freq = AVERAGE(Customer_Demographics[Purchase_Count])
```

This measure calculates the average purchase frequency.

- AVERAGE(Purchase_Count) takes the mean of all customers’ purchase counts.

Why: We use this to understand the overall purchasing frequency across the customer base, which helps identify general buying behaviour.

○ Region

To compare the Average Purchase Frequency by Region, I created a stacked bar chart.

○ Age Group

To compare the Average Purchase Frequency by Age group, I created a clustered column chart.

○ Loyalty Tier

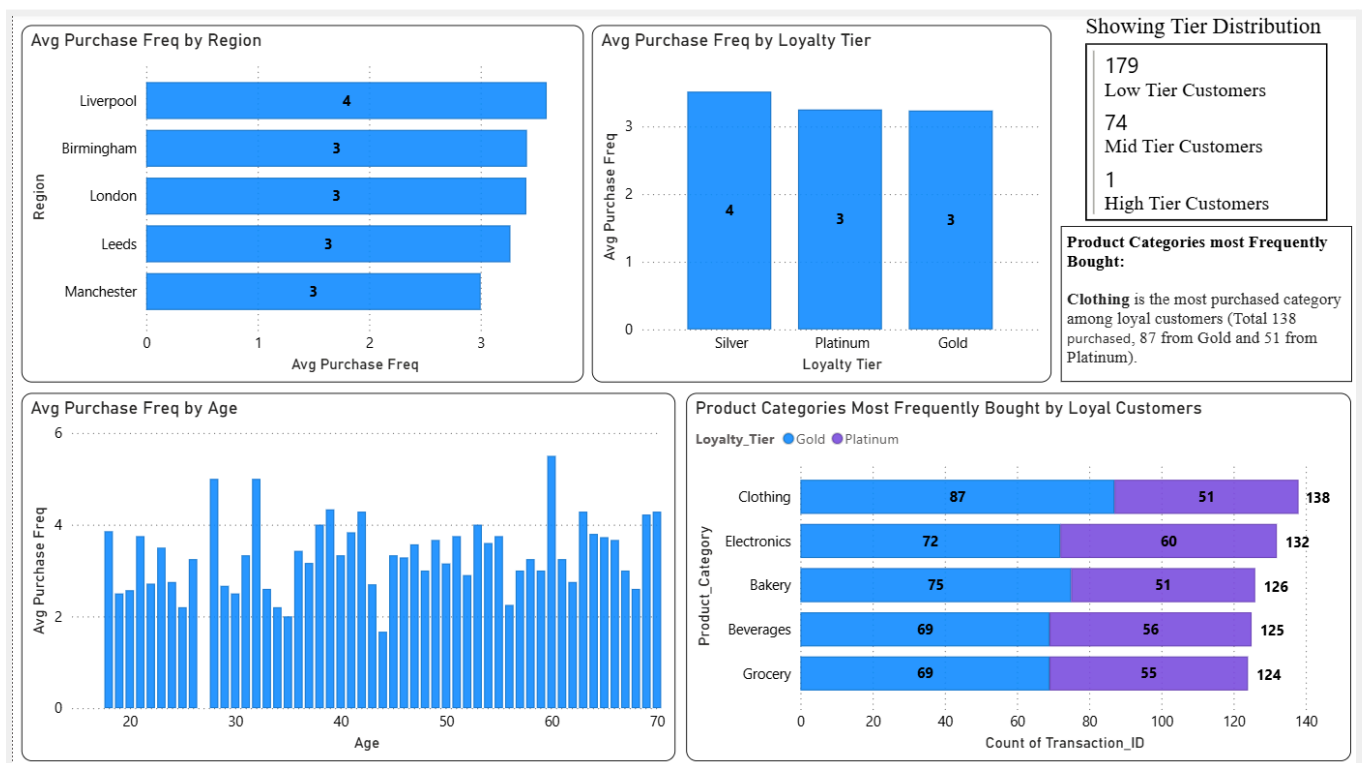
To compare the Average Purchase Frequency by Loyalty Tier, I created a stacked column chart.

➤ Identify product categories most frequently bought by loyal customers

To identify the product categories most frequently purchased by loyal customers, I created a stacked bar chart. On the Y-axis, I placed the product categories, while on the X-axis, I included the count of Transaction IDs. In the legend, I specified the Loyalty Tier and applied a basic filter to include only Gold and Platinum Tier customers.

After visualisation, I have created a text box to indicate which product categories are most frequently bought.

This is the task 3 Report Page view. As we can see in the image:



Interpretation of the page:

★ Overall Customer Tier Distribution

- 179 Low Tier Customers (majority of base)
- 74 Mid Tier Customers
- 1 High Tier Customer

The customer base is heavily skewed towards low-tier customers, with very few reaching higher loyalty tiers. Retention and rewards strategy may need strengthening.

★ Avg Purchase Frequency by Region

- Liverpool leads with the highest avg frequency (4 purchases).
- Leeds follows closely (3.8).
- Birmingham, London, and Manchester lag (3 purchases).

Liverpool and Leeds customers are more engaged, while other regions show lower purchase frequency.

★ Avg Purchase Frequency by Loyalty Tier

- Silver tier customers purchase the most (4 avg purchases).
- Platinum and Gold tiers have slightly lower frequency (3 purchases each).

Surprisingly, lower-tier (Silver) customers are more active than high-tier ones. This may suggest that loyalty benefits aren't strong enough to boost higher-tier activity.

★ Avg Purchase Frequency by Age

- Most customers purchase between 2 and 5 times.
- Spikes: early 30s (5 purchases), age 60 (6 purchases).
- Dips: mid-30s, early 50s (as low as 2 purchases).

Older customers (60 years) and young adults (late 20s – early 30s) show the highest engagement. Middle-aged customers are less frequent buyers.

★ Product Categories Purchased (Gold & Platinum Customers)

- Clothing is the top category (138 transactions total: 87 Gold, 51 Platinum).
- Followed closely by Electronics (132), Bakery (126), Beverages (125), and Grocery (124).

Loyal customers tend to buy Clothing and Electronics the most, showing these are key product drivers for loyalty.

★ Product Categories most Frequently Bought:

Clothing is the most purchased category among loyal customers (Total 138 purchased, 87 from Gold and 51 from Platinum).

Recommendation for TESCO:

1. Strengthen loyalty program: provide stronger incentives to move customers from Silver to Gold/Platinum.
2. Regional targeting: replicate Liverpool & Leeds engagement strategies in Birmingham, London, and Manchester.
3. Age-based offers: design campaigns targeting mid-aged customers (35 - 55) to boost frequency.
4. Double down on top categories (Clothing & Electronics) for promotions and loyalty rewards.

Task 4 Promotion & Loyalty Impact

➤ Create Measure:

○ % of transactions where promotion was applied

To calculate the percentage of Promotional Transactions, I first need to determine the number of Promotional Transactions and the total number of transactions.

To calculate the number of Promotional Transactions, I created a measure using the following DAX formula:

```
Promo Transactions =  
CALCULATE(COUNTROWS(Customer_Transactions),Customer_Transactions[Promotion  
_Applied]="Yes")
```

This measure counts the total number of transactions where a promotion was applied.

- COUNTROWS(Customer_Transactions) counts all transaction records.
- CALCULATE(..., Promotion_Applied = "Yes") filters the data so only promotion-applied transactions are included.

Why: We use this to measure the number of transactions in which a promotion was applied.

Then, I need to calculate the total number of Transactions. I created a measure using the following DAX formula:

```
Number of Transactions = COUNTROWS(Customer_Transactions)
```

This measure counts the total number of transactions recorded.

- COUNTROWS(Customer_Transactions) simply counts all rows in the Customer_Transactions table, with each row representing one transaction.

Why: We use this to know the overall number of transactions.

Afterwards, I created a measure called "Promo Transactions %" using the following DAX formula:

```
Promo Transactions % = DIVIDE([Promo Transactions],[Number of  
Transactions])*100
```

This measure calculates the percentage of transactions made with a promotion.

- DIVIDE([Promo Transactions], [Number of Transactions]) safely divides promo transactions by total transactions.
- Multiplying by 100 converts it into a percentage.

Why: We use this to determine the percentage of transactions where a promotion was applied.

○ Avg. purchase amount with vs without promotion

To calculate the average purchase amount with and without promotions, I created two measures: the first is Average Purchase With Promotion, and the second is Average Purchase Without Promotion.

I have created a measure called "Avg Purchase with Promo" using the following DAX formula:

```
Avg Purchase With Promo =  
CALCULATE(AVERAGE(Customer_Transactions[Amount]),Customer_Transactions[Pro  
motion_Applied]="Yes")
```

This measure calculates the average purchase amount for transactions where a promotion was applied.

- `AVERAGE(Customer_Transactions[Amount])` computes the mean transaction value.
- `CALCULATE(..., Promotion_Applied = "Yes")` filters the data to include only promotion-applied transactions.

Why: We use this to determine the average value of purchases when a promotion is applied.

Then, I have created a measure called "Avg Purchase without Promo" using the following DAX formula:

```
Avg Purchase Without Promo =  
CALCULATE(AVERAGE(Customer_Transactions[Amount]),Customer_Transactions[Pro  
motion_Applied]="No")
```

Why: We use this to determine the average value of purchases when a promotion is not applied.

➤ Compare churn rate across loyalty tiers

To compare the Churn Rate by Loyalty Tiers, I created a stacked column chart.

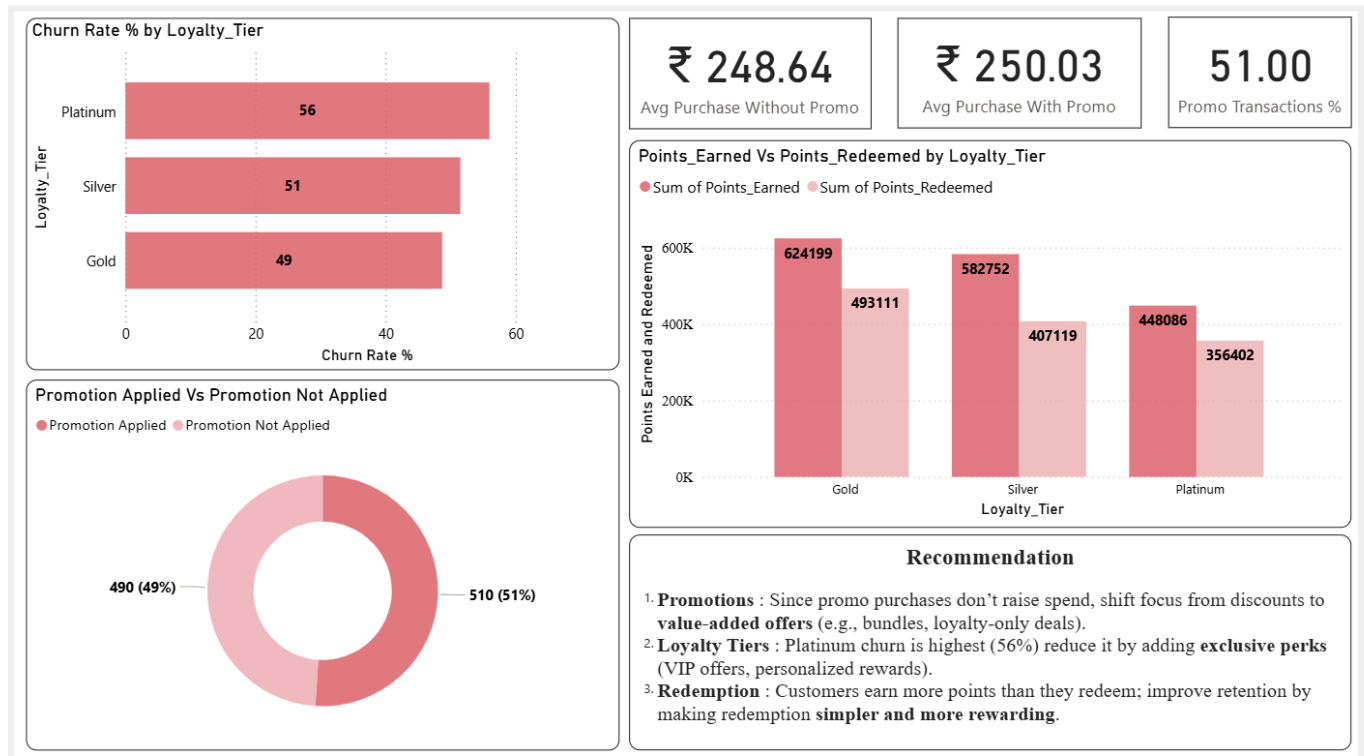
➤ Show Points Earned vs Redeemed by Tier using a clustered column chart

To display Points Earned vs Redeemed by Loyalty Tier, I created a clustered column chart.

➤ Recommend how to improve redemption and retention

I have created a text box that includes three recommendations for TESCO.

This is the task 4 Report Page view. As we can see in the image:



Interpretation of the page:

★ Overall Promotion Effectiveness

- Avg Purchase Without Promo = ₹248.64
- Avg Purchase With Promo = ₹250.03 (almost the same).
- Promo Transactions = 51% (half of purchases).

Promotions don't increase spending significantly. Customers might be habituated to discounts rather than motivated by them.

★ Churn Rate by Loyalty Tier

- Platinum customers churn the most (56%) – this is alarming because they are supposed to be the most loyal.
- Silver churn = 51%, Gold churn = 49%.

Even the highest loyalty tier is not providing enough stickiness, signalling weak differentiation in rewards.

★ **Points Earned vs Redeemed by Loyalty Tiers**

- Gold tier: Earned 624,199 vs Redeemed 493,111 - 79% redemption
- Silver tier: Earned 582,752 vs Redeemed 407,119 - 70% redemption.
- Platinum tier: Earned 448,086 vs Redeemed 356,402 - 79% redemption.

Across tiers, a large chunk of points remains unused, especially in Silver. This weakens the value perception of the loyalty program.

★ **Promotion Applied vs Not Applied**

- 51% purchased used promotions, 49% did not.

Balanced split, but since promos don't lift spending, they're not driving incremental value.

Recommendation for TESCO:

1. Promotions: Since promo purchases don't raise spend, shift focus from discounts to value-added offers (e.g., bundles, loyalty-only deals).
2. Loyalty Tiers: Platinum churn is highest (56%), reduce it by adding exclusive perks (VIP offers, personalised rewards).
3. Redemption: Customers earn more points than they redeem; improve retention by making redemption simpler and more rewarding.

Task 5 Store Performance vs Retention

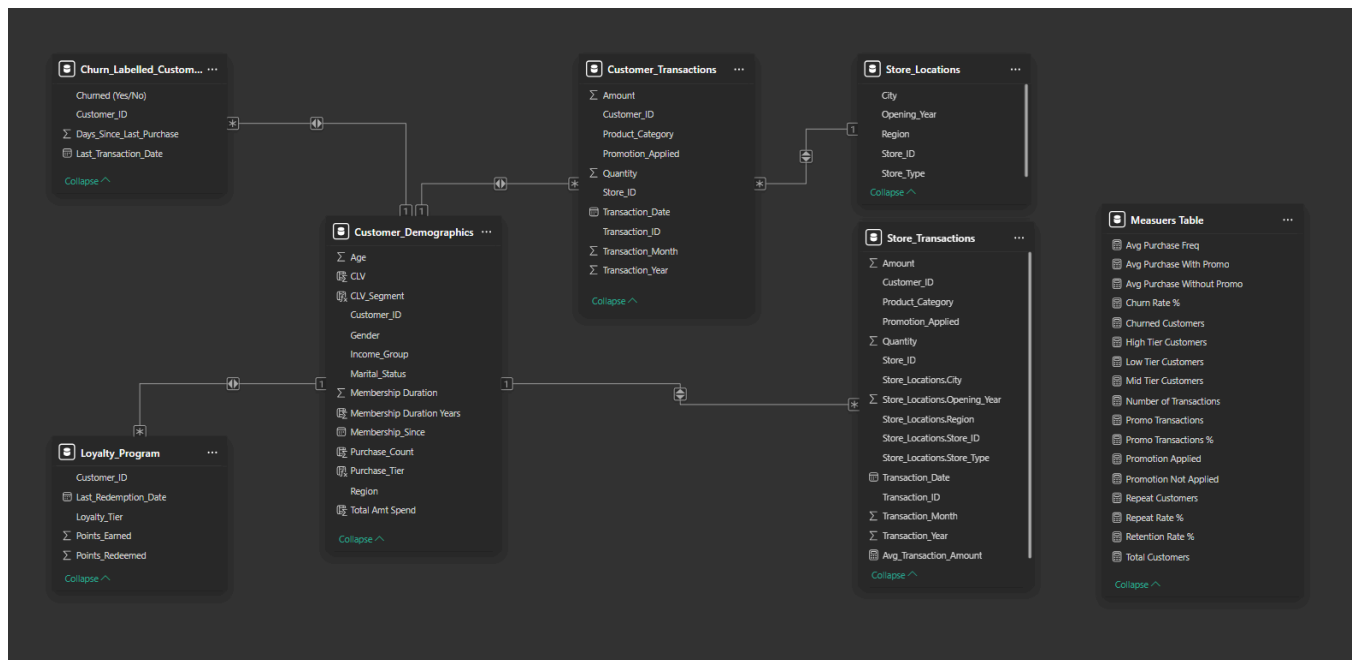
➤ Merge Store_ID from transactions with Store_Locations

To merge the Store_ID from the Transaction table with the Store_Location table, I used the Merge Queries function. First, I opened Power Query and selected the Transaction table. Under the Home tab, I chose the Merge Queries function and selected "Merge Queries as New" because I wanted to create a new merged table.

In the Merge box, I selected the Store_Location table as a second table and chose the Store_ID column from both tables. For the type of join, I selected "Full Outer" and clicked OK. I then renamed the new table as Store_Transaction.

After that, in the Model View, I created a one-to-many relationship between the Customer_Demographics table and the Store_Transaction table.

As we can see in the image below:



➤ Visualize:

○ Avg. transaction amount by store type

To calculate the Average Transaction Amount, I created a measure called Avg Transaction Amount using this DAX formula:

```
Avg_Transaction_Amount = AVERAGE(Store_Transactions[Amount])
```

This measure calculates the average value of all transactions.

- `AVERAGE(Store_Transactions[Amount])` takes the mean of the transaction amounts from the Store_Transactions table.

Why: We use this to calculate the Average Transaction Amount.

And, to display the Average Transaction Amount by Store Type, I created a Doughnut chart.

○ Churn rate by store region

To compare the Churn Rate by Store Region, I created a stacked column chart.

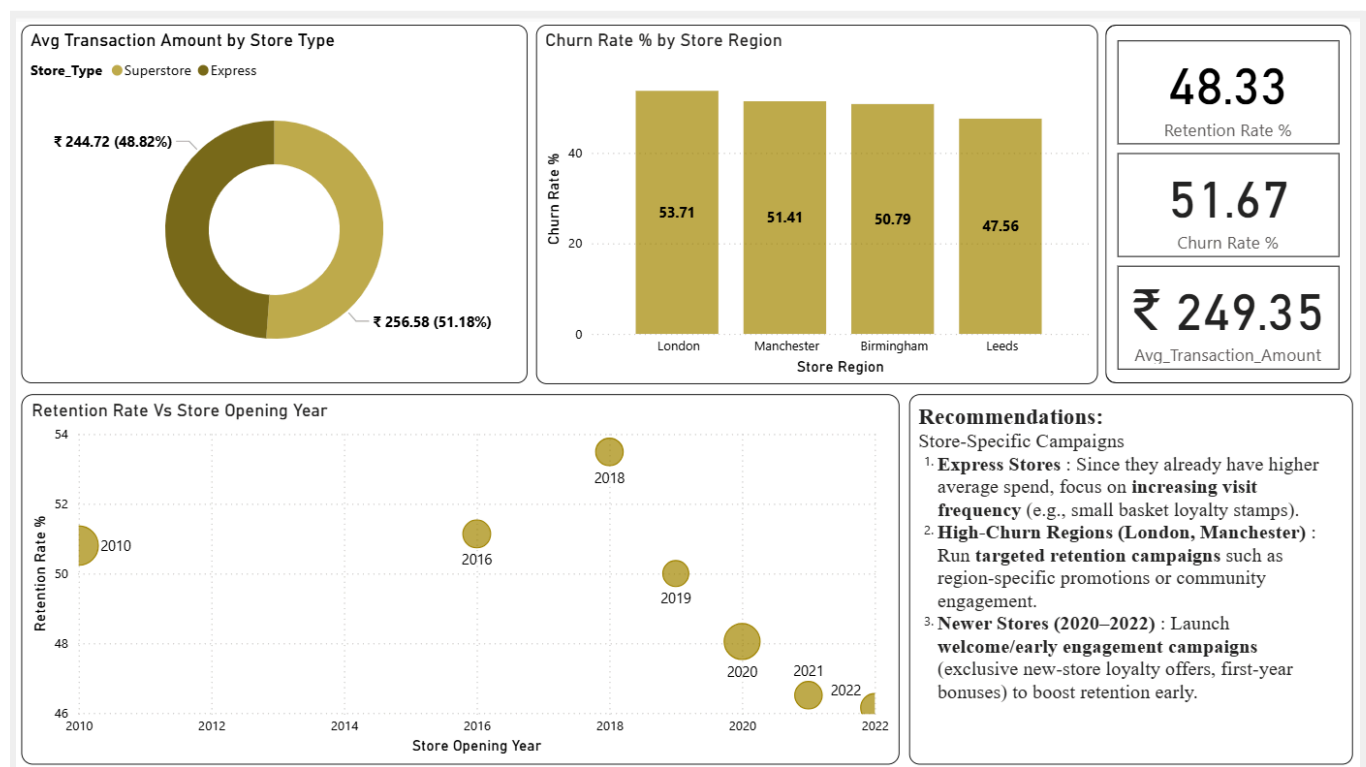
○ Correlation between store opening year and retention

To correlate the store opening year and the retention rate, I created a scatter plot in which I included the total number of customers in the size dimension.

➤ Suggest where to run store-specific campaigns

I have created a text box that includes three recommendations for TESCO regarding store-specific campaigns.

This is the task 5 Report Page view. As we can see in the image:



Interpretation of the page:

★ Overall KPIs

- Retention Rate: 48.33%
- Churn Rate: 51.67%
- Avg Transaction Amount: ₹249.35

Just under half of customers are retained, showing room for stronger engagement strategies.

★ Avg Transaction Amount by Store Type

- Express Stores: ₹256.58 (51.18%)
- Superstores: ₹244.72 (48.82%)

Express stores generate slightly higher transaction values per visit, even though they are smaller-format.

★ Churn Rate by Store Region

- Highest churn: London (53.71%) and Manchester (51.41%)
- Lower churn: Birmingham (50.79%), Leeds (47.56%)

London and Manchester stores are struggling more with retention compared to Leeds.

★ Retention Rate vs Store Opening Year

- Older stores (2010, 2016, 2018): Higher retention (50–53%).
- Newer stores (2020–2022): Lower retention (below 48%, some as low as 46%).

Newer stores face challenges in building early loyalty, while older stores maintain stronger customer stickiness.

Recommendation for TESCO:

1. Express Stores: Since they already have a higher average spend, focus on increasing visit frequency (e.g., small basket loyalty stamps).
2. High-Churn Regions (London, Manchester): Run targeted retention campaigns such as region-specific promotions or community engagement.
3. Newer Stores (2020–2022): Launch welcome/early engagement campaigns (exclusive new-store loyalty offers, first-year bonuses) to boost retention early.

Task 6 Customer Value (CLV) Analysis

➤ Calculate CLV = Total Amount Spent / Membership Duration (in years)

To calculate Customer Lifetime Value (CLV), I need to determine the Total Amount Spent and Membership Duration in years. Therefore, I have created calculated columns to compute CLV for each customer.

To calculate the Total Amount spent for each customer, I created a Calculated Column as Total Amt Spent, using this DAX formula:

```
Total Amt Spend =
VAR Total_Amt_Spend =
    CALCULATE(
        SUM(Customer_Transactions[Amount]),
        FILTER(Customer_Transactions, Customer_Transactions[Customer_ID] =
            Customer_Demographics[Customer_ID])
    )
RETURN
IF(ISBLANK(Total_Amt_Spend), 0, Total_Amt_Spend)
```

This calculated column shows the total amount each customer has spent.

- SUM(Customer_Transactions[Amount]) adds up all transaction values.
- FILTER(... Customer_ID match ...) ensures only the transactions belonging to the current customer are included.
- VAR Total_Amt_Spend stores this sum.
- IF(ISBLANK(...), 0, Total_Amt_Spend) replaces blanks with 0 for customers who haven't made any purchases.

Why: We use this to assign a spending total to each customer, which is useful for customer value segmentation and revenue analysis.

To calculate the Membership Duration in years for each customer, I converted the Membership Duration column, which is in days, into years. I created a calculated column called Membership Duration Years using this DAX formula:

```
Membership Duration Years =
DATEDIFF(Customer_Demographics[Membership_Since], TODAY(), YEAR)
```

This calculated column measures how long a customer has been a member, expressed in years.

- DATEDIFF(Membership_Since, TODAY(), YEAR) calculates the difference between the

customer's membership start date and today's date, in years.
Why: We use this to easily analyse customer tenure in years.

Afterwards, I have calculated the CLV for each customer by using this DAX formula.

```
CLV = DIVIDE(Customer_Demographics[Total Amt Spend],Customer_Demographics[Membership Duration Years],0)
```

This calculated column estimates a customer's average spend per year.

- Total Amt Spend gives the total amount spent by the customer.
- Membership Duration Years shows how long they've been a member (in years).
- `DIVIDE(..., ..., 0)` divides spend by duration, with 0 as the fallback if the denominator is blank or zero.

Why: We use this method to calculate the Customer Lifetime Value (CLV) for each customer.

As we can see in the image below:

1 CLV = DIVIDE(Customer_Demographics[Total Amt Spend],Customer_Demographics[Membership Duration Years],0)												
Gender	Age	Membership Since	Marital Status	Region	Income_Group	Membership Duration	Purchase_Count	Purchase_Tier	Membership Duration Years	Total Amt Spend	CLV	CLV_Segme
Male	50	01 November 2020	Single	London	High	1776	3	Low	5	₹ 305.31	₹ 61.06	Low
Male	36	18 August 2021	Single	Birmingham	Medium	1486	5	Mid	4	₹ 1,092.27	₹ 273.07	Medium
Female	61	19 November 2020	Single	Liverpool	High	1758	0	New	5	₹ 0.00	₹ 0.00	Low
Male	21	03 May 2023	Single	Manchester	Medium	863	4	Low	2	₹ 1,010.38	₹ 505.19	High
Female	51	11 July 2023	Single	London	High	794	2	Low	2	₹ 141.38	₹ 70.69	Low
Female	23	21 March 2024	Single	Liverpool	Low	540	3	Low	1	₹ 522.34	₹ 522.34	High
Male	39	07 June 2023	Single	Leeds	Low	828	4	Low	2	₹ 513.73	₹ 256.87	Medium
Female	33	19 December 2022	Single	London	Low	998	1	One-time	3	₹ 263.37	₹ 87.79	Low
Female	33	24 November 2023	Single	Birmingham	Medium	658	5	Mid	2	₹ 1,346.28	₹ 673.14	High
Female	46	08 September 2021	Single	Birmingham	High	1465	2	Low	4	₹ 158.67	₹ 39.67	Low
Female	20	30 August 2021	Single	Liverpool	Low	1474	5	Mid	4	₹ 1,032.67	₹ 258.17	Medium
Male	36	26 November 2023	Single	Manchester	Medium	656	2	Low	2	₹ 855.22	₹ 427.61	Medium
Female	20	02 August 2022	Single	Birmingham	High	1137	2	Low	3	₹ 654.42	₹ 218.14	Medium
Female	36	26 August 2023	Single	Birmingham	Low	748	2	Low	2	₹ 363.27	₹ 182.64	Medium
Male	24	05 April 2023	Single	London	Low	891	3	Low	2	₹ 741.20	₹ 370.60	Medium
Male	69	10 February 2024	Single	London	Medium	580	7	Mid	1	₹ 2,048.97	₹ 2,048.97	High
Female	56	26 February 2024	Single	London	Low	564	3	Low	1	₹ 822.69	₹ 822.69	High
Male	45	08 October 2023	Single	London	High	705	1	One-time	2	₹ 62.65	₹ 31.33	Low
Female	67	14 September 2023	Single	London	Low	729	5	Mid	2	₹ 1,379.98	₹ 689.99	High
Female	33	10 September 2021	Single	Leeds	Low	1463	0	New	4	₹ 0.00	₹ 0.00	Low
Male	69	30 June 2022	Single	Birmingham	High	1170	5	Mid	3	₹ 920.76	₹ 306.92	Medium
Male	19	29 June 2021	Single	London	Low	1536	4	Low	4	₹ 1,226.89	₹ 306.72	Medium
Male	65	26 December 2022	Single	Liverpool	High	991	2	Low	3	₹ 168.33	₹ 56.11	Low
Male	29	07 September 2023	Single	Liverpool	Medium	736	3	Low	2	₹ 574.74	₹ 287.37	Medium
Female	22	01 October 2022	Single	London	Medium	1077	4	Low	3	₹ 709.19	₹ 236.40	Medium
Female	54	04 August 2020	Single	Birmingham	High	1865	2	Low	5	₹ 760.94	₹ 152.19	Low
Male	49	10 May 2024	Single	Birmingham	Medium	490	4	Low	1	₹ 798.66	₹ 798.66	High
Female	58	21 November 2023	Single	Leeds	Medium	661	1	One-time	2	₹ 304.71	₹ 152.36	Low

➤ Segment customers into:

- Low (Bottom 25%)
- Medium (Mid 50%)
- High (Top 25%)

To create customer segments categorised as Low, Medium, and High based on the above conditions, I created a calculated column named CVL segment next to CVL using this DAX formula:

```
CLV_Segment =
VAR CLV_Value = Customer_Demographics[CLV]
VAR P25 = PERCENTILEX.INC(Customer_Demographics,
Customer_Demographics[CLV], 0.25)
VAR P75 = PERCENTILEX.INC(Customer_Demographics,
Customer_Demographics[CLV], 0.75)
RETURN
    SWITCH(
        TRUE(),
        CLV_Value <= P25, "Low",
        CLV_Value >= P25 && CLV_Value <= P75, "Medium",
        "High"
    )
```

This calculated column classifies customers into Low, Medium, or High segments based on their CLV.

- CLV_Value stores the customer's CLV.
- P25 and P75 calculate the 25th and 75th percentiles of CLV across all customers.
- SWITCH(TRUE(), ...) then assigns:
 - Low - CLV below or equal to the 25th percentile.
 - Medium - CLV between the 25th and 75th percentile.
 - High - CLV above the 75th percentile.

Why: We use this to segment customers by lifetime value, which helps prioritise retention for high-value customers.

➤ Visualize:

○ CLV vs Days Since Last Purchase

To display Customer Lifetime Value (CLV) versus Days Since Last Purchase, I created a scatter plot and included the CLV segment in the legend to identify high-value customers.

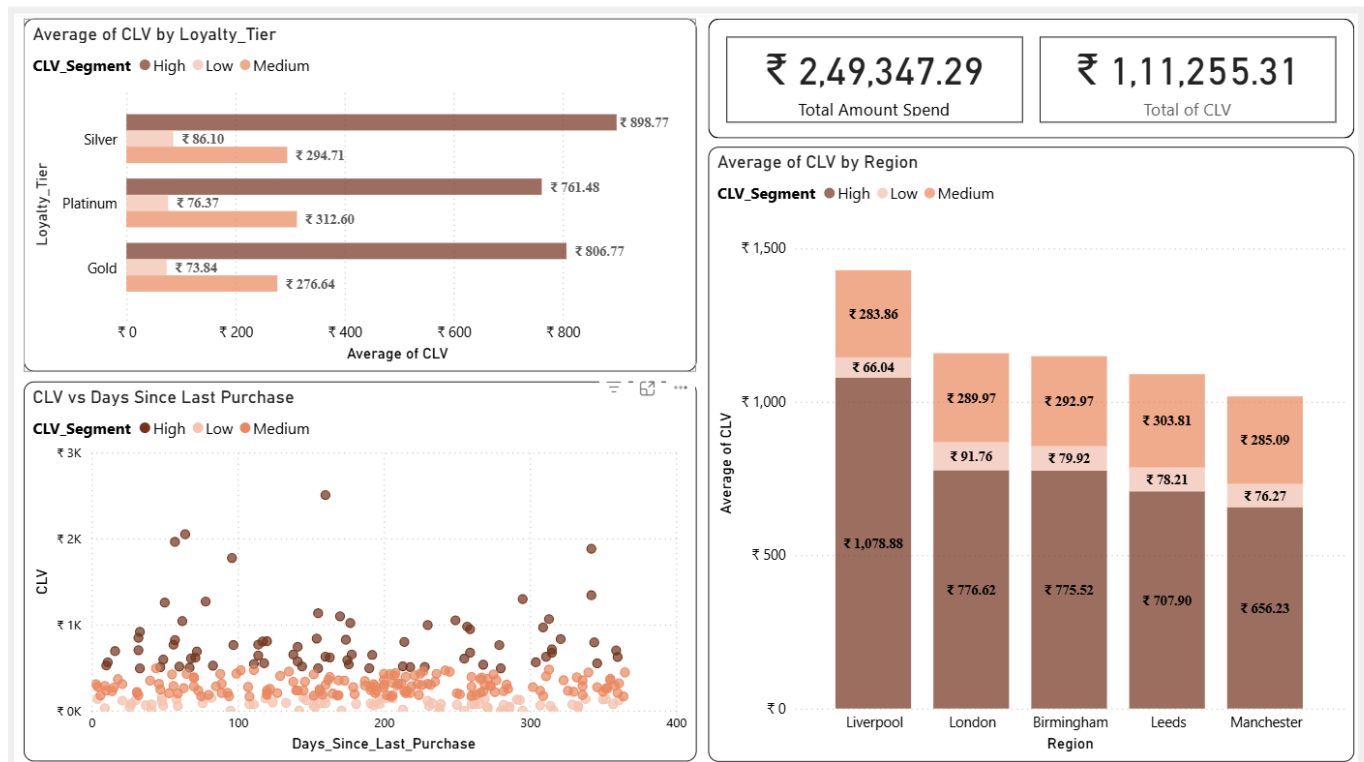
○ CLV by Loyalty Tier and Region

I have created two separate charts.

To compare and visualise CLV by Loyalty Tier, I created a clustered bar chart and included the CLV segment in the legend to identify high-value customers.

To compare and visualise CLV by Region, I created a clustered bar chart and included the CLV segment in the legend to identify high-value customers.

This is the task 6 Report Page view. As we can see in the image:



Interpretation of the page:

★ Overall Metrics

- Total Amount Spent: ₹2,49,347.29
- Total CLV (Customer Lifetime Value): ₹1,11,255.31

CLV is less than half of the total spend, suggesting many purchases are one-offs or customers churn before realising higher lifetime value.

★ Average CLV by Loyalty Tier

- Silver customers have the highest CLV (₹898.77 High segment).
- Gold follows (₹806.77 High segment).
- Platinum lags (₹761.48 High segment) despite being the top tier.

Silver customers deliver the best long-term value, which is unusual — loyalty upgrades (Gold/Platinum) are not translating into higher CLV.

★ **Average CLV by Region**

- Liverpool leads strongly (₹1,078.88 High segment) – much higher than any other region.
- Other regions:
- London: ₹776.62
- Birmingham: ₹775.52
- Leeds: ₹707.90
- Manchester: ₹656.23 (lowest).

Liverpool customers are the most valuable, while Manchester contributes the least to CLV.

★ **CLV vs Days Since Last Purchase**

- High CLV customers are spread across time – some haven't purchased recently, showing risk of attrition.
- Many low/medium CLV customers cluster around recent purchases, but their spend is small.

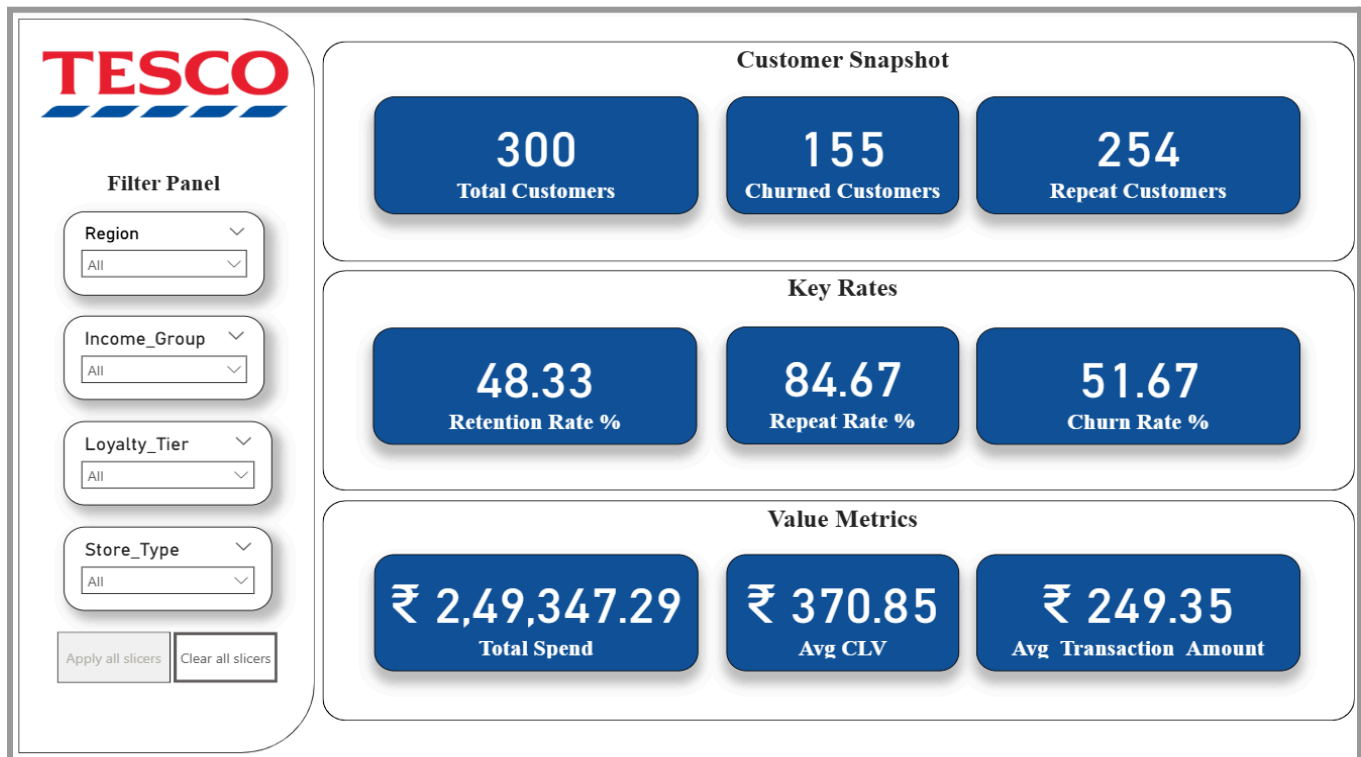
Even valuable customers (high CLV) can go dormant if not engaged. Timing of reactivation campaigns is key.

Recommendation for TESCO:

1. Rebalance loyalty program – strengthen benefits in Gold/Platinum tiers to justify upgrades and increase their CLV.
2. Prioritise Liverpool as a growth model – replicate engagement strategies from Liverpool in other regions, especially Manchester.
3. Reactivation campaigns for high-CLV dormant customers – use personalised offers, reminders, or exclusives to bring them back.

Task 7 Final Dashboard and Executive Summary

- Create a multi-page Power BI report:
 - Page 1: Overview KPIs (Churn, CLV, Repeat Rate)



Interpretation of the page:

★ Customer Snapshot

- Total Customers (300) - The base population being analysed.
- Churned Customers (155) - How many customers stopped purchasing?
- Repeat Customers (254) - How many are still active & loyal - strength of retention.

Together, these 3 show: Customer base health, who stayed vs. who left. Although repeat purchases appear strong, the churn rate remains high, suggesting that many customers depart after initial interactions.

★ Key Rates

- Retention Rate % (48.33%) - % of customers retained. Indicates the success of loyalty strategies.
- Repeat Rate % (84.67%) - % of customers making multiple purchases, engagement level.
- Churn Rate % (51.67%) - % of customers lost. A critical metric for long-term growth.

These 3 show: Customer behaviour & risk levels, The business effectively drives repeat purchases once customers are engaged, but it struggles with initial retention, resulting in too many customers dropping off early.

★ **Value Metrics**

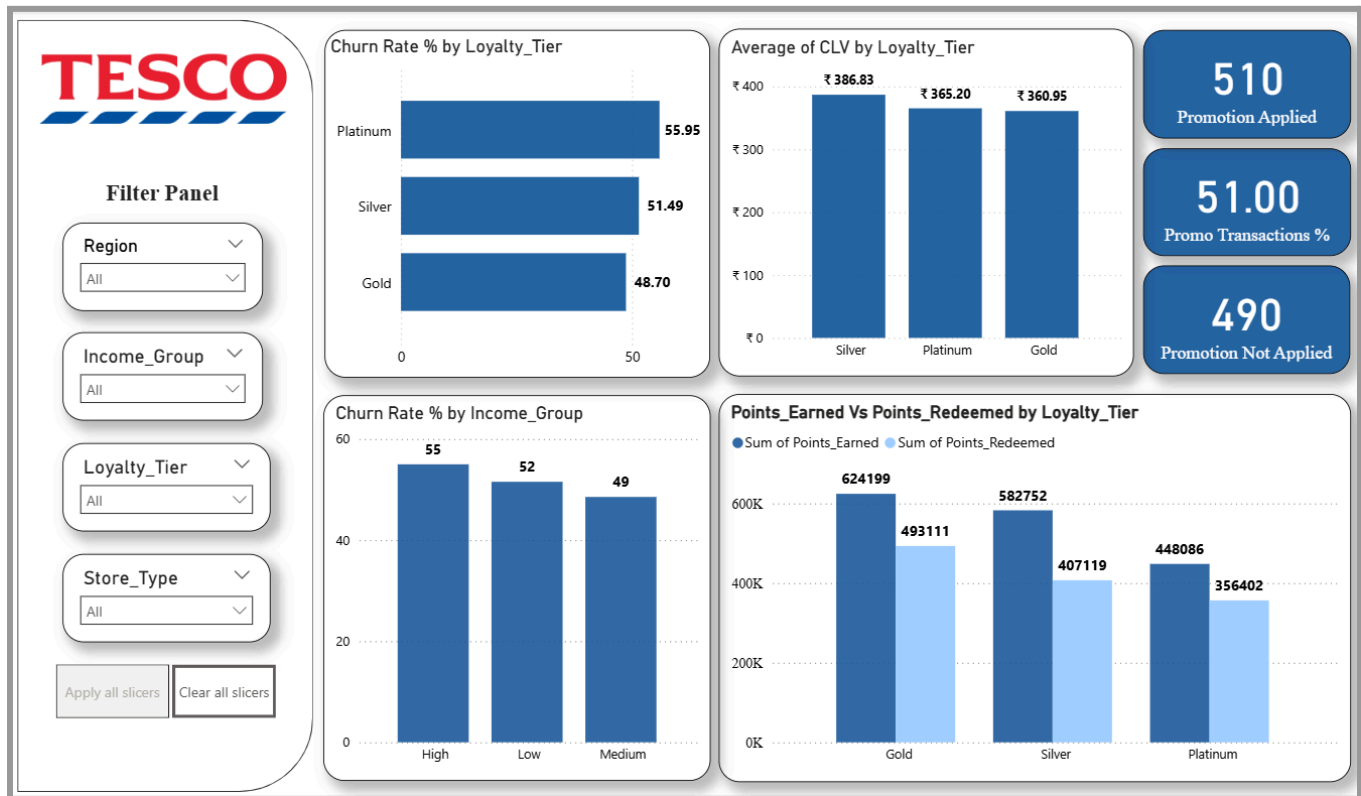
- Total Spend (₹2,49,347.29) - Overall revenue contribution from the customer base.
- Average CLV (₹370.85) - Profitability per customer over time. Shows long-term value.
- Avg Transaction Amount (₹249.35) - Typical basket size, helps spot spending patterns.

These 3 show: How much value customers are driving. Customers who remain engaged generate higher lifetime value, indicating that repeat transactions significantly boost profitability. However, low retention negatively impacts the overall potential for growth.

★ **Filter Panel (on the left side)**

- Region: Lets you slice data by geographic areas (Liverpool, London, Birmingham, etc.) to see regional retention, churn, and spend differences.
- Income Group: Allows analysis of customers by income segment, helping spot whether high/low-income groups behave differently.
- Loyalty Tier: Filters by loyalty levels (Silver, Gold, Platinum), useful to see how higher tiers impact CLV, repeat rate, or churn.
- Store Type: Splits analysis between Express vs Superstores, showing differences in average spend and retention.

○ Page 2: Loyalty & Promotion Impact



Interpretation of the page:

★ KPI Cards (Top-Right)

- Promotion Applied (510)
- Why I use: Quick snapshot of how many transactions are involved in promotions.
- Insight: The scale of promotion adoption. A higher number means promotions are a major lever for sales
- Promo Transactions % (51%)
- Why I use: Percentages are more intuitive than raw counts for comparison.
- Insight: Just over half of all transactions were promotion-based, with heavy reliance on promotions.
- Promotion Not Applied (490)
- Why I use: Completes the picture (applied vs not applied).
- Insight: Almost half of transactions were without promotions, scope to analyse if non-promo purchases generate more loyalty/profit.

★ Churn Rate % by Loyalty Tier

- Why I use: Helps pinpoint which loyalty tier is struggling most with retention.
- Platinum customers surprisingly have the highest churn rate (55.95%), despite being premium.
- Silver customers churn at 51.49%, and Gold customers have the lowest churn (48.70%).

Insight: Loyalty tier alone doesn't guarantee retention; Platinum customers may expect stronger benefits or exclusive engagement.

★ Average CLV by Loyalty Tier

- Why I use: Shows the financial value of customers in each tier
- Silver (₹386.83), Platinum (₹365.20), and Gold (₹360.95) are fairly close in terms of CLV.
- Despite lower churn, Gold CLV is slightly lower, which suggests they are retained but spend less per customer.

Insight: Silver customers show the best balance of spend and loyalty, while Platinum churn is a concern.

★ Churn Rate % by Income Group

- Why we use: To compare churn risk across income levels.
- High-income customers churn the most (55%), followed by low-income (52%), and medium-income churn is the lowest (49%).

Insight: Medium-income customers are more stable; high-income groups may be less price-sensitive but demand stronger value/experience.

★ Points Earned vs Points Redeemed by Loyalty Tier

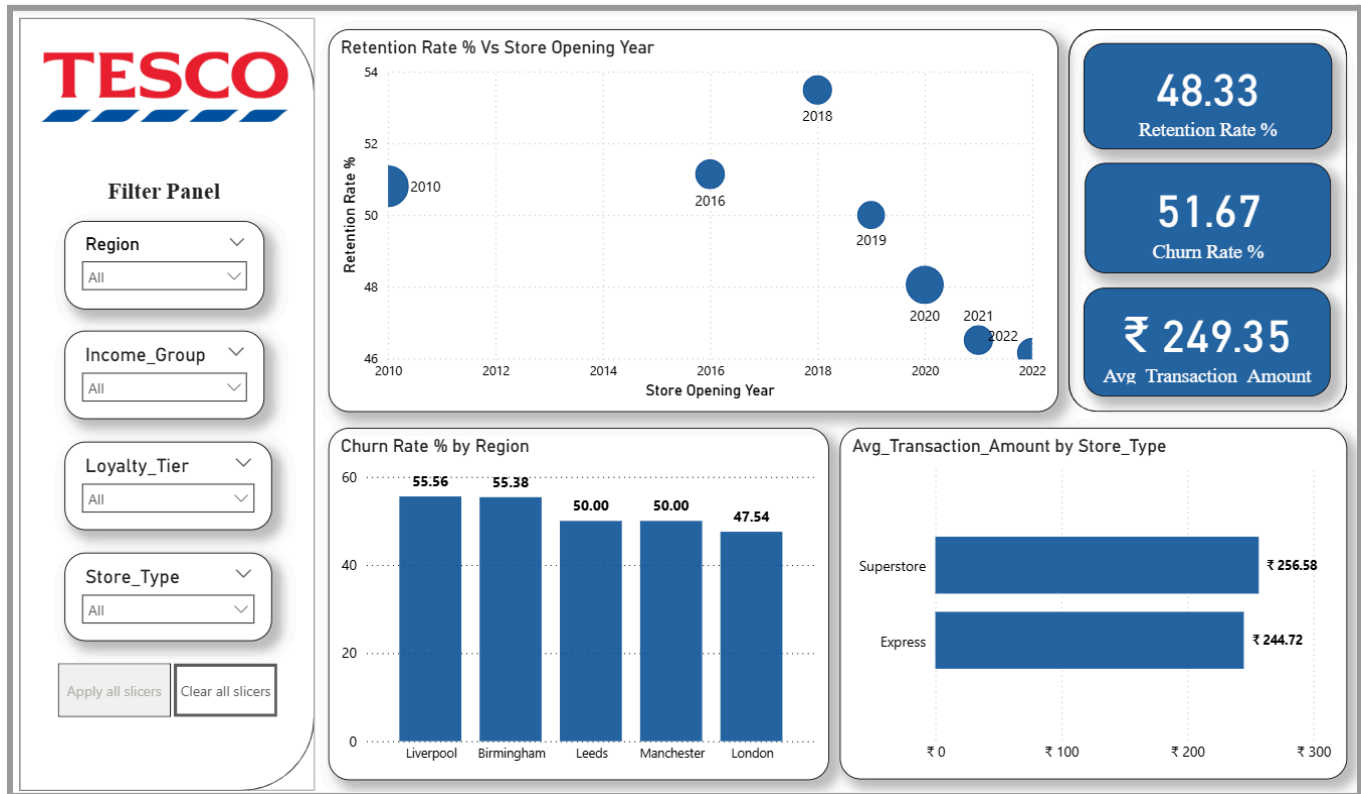
- Why we use: Evaluates the effectiveness of the loyalty program.
- Gold Tier: 624K points earned, 493K redeemed - 79% redemption (good engagement).
- Silver Tier: 582K earned, 407K redeemed - 70% redemption.
- Platinum Tier: 448K earned, 356K redeemed - 79% redemption.

Insight: Redemption rates are strong across tiers, with Gold & Platinum customers showing higher engagement with loyalty programs.

★ Filter Panel (Left Side)

- Region: Compare how loyalty and promotions vary by geography (e.g., does churn differ in London vs Birmingham?).
- Income Group: Drill down into specific customer groups to see spending/churn behaviour by income.
- Loyalty Tier: Filter data to focus on Platinum, Gold, or Silver exclusively.
- Store Type: Compare Express vs Superstores to see if promotion impact differs by format.

○ Page 3: Store and Region Insights



Interpretation of the page:

★ KPI Cards (Top-Right)

- Retention Rate % (48.33%)
- Why I use: A direct measure of how well Tesco is keeping its customers.
- Insight: Out of all customers, only 48% are staying (retained). This is a red flag that retention is weak and needs targeted campaigns.
- Churn Rate % (51.67%)
- Why I use: Complements retention by showing loss. Executives care about churn because it signals revenue leakage.
- Insight: More than half of customers are leaving - a critical insight for prioritising retention efforts.
- Average Transaction Amount (₹249.35)
- Why I use: Provides a quick revenue perspective per transaction.
- Insight: On average, each purchase contributes ₹249, helping link customer behaviour with store performance.

★ Retention Rate % vs Store Opening Year

- Why I use: Highlights whether store age impacts retention.
- Older stores (2010, 2016, 2018) - higher retention (50 - 54%).
- Newer stores (2020–2022) - lower retention (46 - 48%).

Insight: Customers are more loyal in older stores, while newer stores struggle with retention → early engagement strategies are needed.

★ Churn Rate % by Region

- Why I use: A regional comparison is critical for targeting campaigns effectively.
- Highest churn: Liverpool (55.56%) and Birmingham (55.38%).
- Lowest churn: London (47.54%).

Insight: Certain regions (Liverpool & Birmingham) face loyalty issues, possibly due to competition or weak local engagement. London, despite being a big city, is performing relatively better.

★ Average Transaction Amount by Store Type

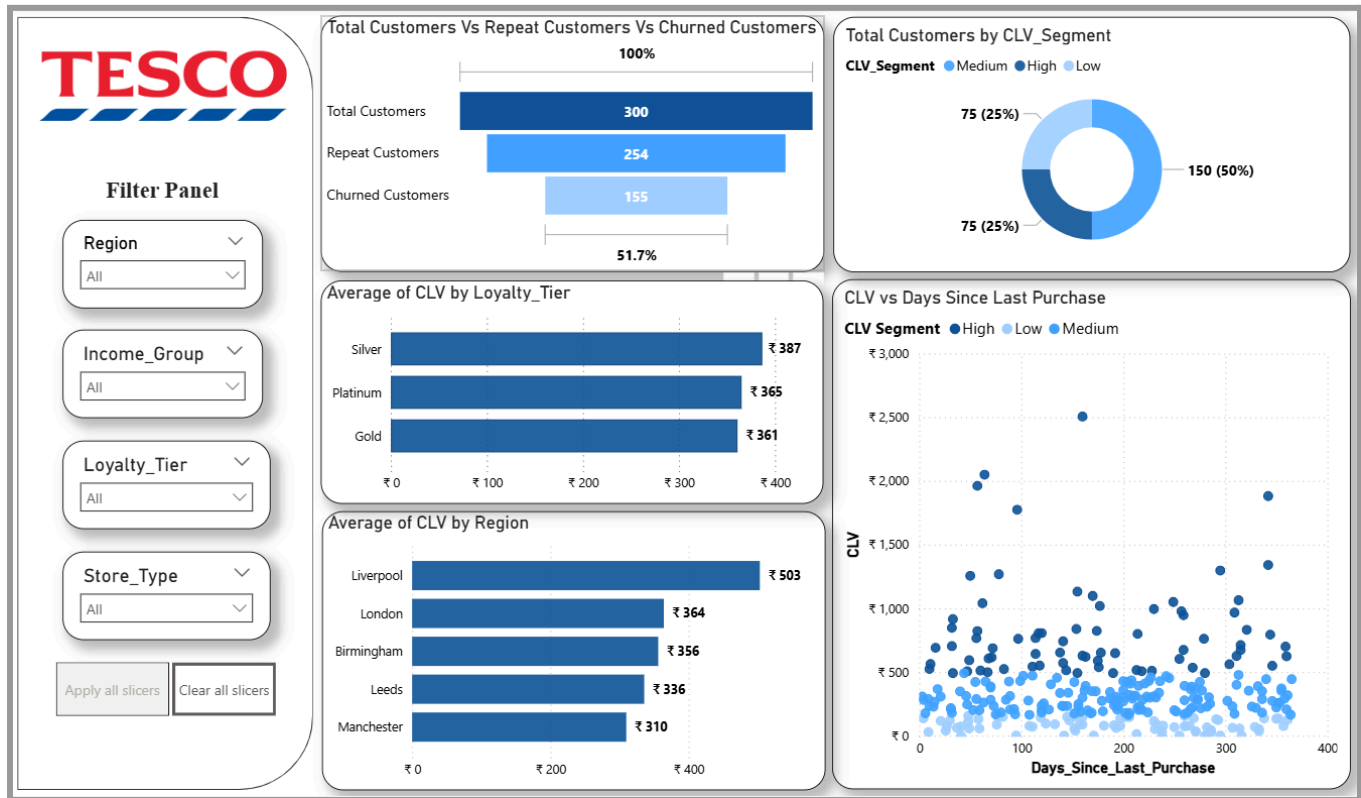
- Why I use: The store type (Express vs. Superstore) directly impacts strategy.
- Superstore: ₹256.58 (higher spend per transaction).
- Express: ₹244.72.

Insight: Superstores drive bigger basket sizes, while Express stores encourage smaller but possibly more frequent visits.

★ Filter Panel (Left Side)

- Region: Drill down by city/region to see retention, churn, and spend differences.
- Income Group: Understand if churn/spending patterns differ by customer income levels.
- Loyalty Tier: Focus on Silver/Gold/Platinum customers for tier-specific insights.
- Store Type: Compare Superstores vs Express stores to see how format affects retention and spend.

○ Page 4: Customer Segmentation (Churned, Repeat, High-Value)



Interpretation of the page:

★ Total Customers vs Repeat Customers vs Churned Customers

- Why I use: Gives a clear breakdown of customer distribution
- Total Customers: 300
- Repeat Customers: 254 (85%)
- Churned Customers: 155 (52%)

Insight: Although a large majority of customers made repeat purchases, churn is still high (over half). This means retention efforts are not sustaining customers long-term.

★ Total Customers by CLV Segment

- Why I use: Doughnut is good for showing proportional splits.
- High CLV: 75 (25%)
- Medium CLV: 150 (50%)
- Low CLV: 75 (25%)

Insight: Half of the customer base is medium-value, while high-value customers are limited to 1 in 4. This indicates potential to push medium customers upward into high-value through targeted loyalty schemes.

★ Average CLV by Loyalty Tier

- Why I use: Allows comparison across loyalty groups (Gold, Platinum, Silver).
- Silver: ₹387 (highest CLV)
- Platinum: ₹365
- Gold: ₹361

Insight: Surprisingly, Silver customers generate the highest CLV, possibly due to frequent mid-ticket purchases. Platinum/Gold tiers may not be delivering expected value - loyalty benefits may need rethinking.

★ Average CLV by Region

- Why I use: Easy to compare customer lifetime value across geographies.
- Highest CLV: Liverpool (₹503)
- Lowest CLV: Manchester (₹310)
- Insight: Liverpool customers spend significantly more, while Manchester lags. Region-specific campaigns can uplift underperforming areas.

★ CLV vs Days Since Last Purchase

- Why I use: Captures two dimensions at once (recency + value).
- High CLV customers tend to cluster closer to recent purchase dates.
- Low/Medium CLV customers are spread across a higher days since last purchase - meaning churn risk is higher here.

Insight: Recent engagement strongly correlates with higher CLV. Focusing on reactivation campaigns for customers who haven't purchased in 100+ days could reduce churn.

★ Filter Panel (Left Side)

- Region: Drill down to see CLV/churn by city.
- Income Group: Identify how spending/loyalty differs across income levels.
- Loyalty Tier: Compare CLV distribution by Silver, Gold, Platinum.
- Store Type: Check whether Superstore or Express influences CLV.
- Apply / Clear All Slicers: Makes the dashboard fully interactive, helping management zoom into specific customer segments.

➤ Create a card or textbox summarising the top 3 recommendations:

- What should TESCO do to retain more customers?
- Where should they focus next?

To present the Executive Summary, Recommendations, and Focus Areas, I created a new page named "Executive Summary" which includes three recommendations for TESCO.

○ Page 5: Executive Summary



Executive Summary - Recommendations & Focus Areas

What TESCO Should Do to Retain More Customers	Where TESCO Should Focus Next
1.Reduce Churn in Platinum Tier: Platinum customers have the highest churn rate (56%), even though they are the most valuable. TESCO should provide exclusive benefits such as VIP-only promotions, early access to sales, and personalized rewards to keep them engaged. 2.Improve Points Redemption Customers are earning more points than they redeem, meaning loyalty value is being left unused. Simplify the redemption process (e.g., one-click use at checkout, small denomination rewards) and make rewards more attractive (bundles, personalized vouchers). 3.Rethink Promotions Promotional purchases do not significantly increase average spend. Instead of focusing only on discounts, shift toward value-added offers like product bundles, free delivery perks, and loyalty-only deals that encourage repeat visits without cutting margins.	1.Platinum Loyalty Segment Retaining high-value Platinum customers should be the top priority. Losing them impacts both revenue and long-term brand loyalty. 2.High-Churn Regions (Liverpool & Birmingham) Both regions show churn above 55%, signaling a need for region-specific retention campaigns, such as local promotions, community events, or tailored marketing. 3.Younger Stores (Opened 2020–2022) Newer stores show lower retention rates. Launch welcome campaigns (exclusive first-year offers, early engagement bonuses) to build habits early and prevent drop-offs.

I have summarised my TESCO Customer Retention Analytics using Power BI.

THANK YOU